Timothy Bernasch

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Profile

Computer engineer with 15+ years building systems end-to-end - from embedded firmware and PCB design to desktop C++ applications and full-stack web platforms. Proven track record modernizing legacy codebases, architecting cross-platform systems, and bridging hardware and software disciplines from PCB design through production deployment.

Experience

Note: My TrackGen Solutions and Dexlabs roles overlap (2016–2026) - these were concurrent positions, and both employers were aware of the arrangement.

Senior Software Engineer — TrackGen Solutions Inc. (trackgen.com)

2010 – 2026

- Developed new features and maintained a large desktop GUI C++ application with 30+ years of legacy code. Collaborated with stakeholders on adding features, fixing bugs, and optimizing performance
- Ported a large desktop GUI C++ application from Windows to Linux, adapting platform-specific components and resolving compatibility issues across the application stack
- Created a prototype of a full web frontend (“Software as a Service”) for a desktop application using Node.js, React, Docker Swarm and AWS services
- Built and maintained a custom build system and release tooling (CMake, shell scripting, Lua, GCC toolchain, MSVC toolchain) to compile, package, and deploy a large application. Developed automation and tooling scripts (Python, Lua, shell scripting) for build systems and internal developer productivity workflows
- Developed an extensive SQL schema to store streaming data efficiently, created normalized tables with foreign key relationships, indexes, and performed optimization of queries on million-row tables
- Implemented and maintained WebSocket communication layer between C++ application and web frontends (Node.JS, React, CesiumJS), for visualization of data inside the application
- Developed embedded Python scripting interface in C++, enabling runtime automation
- Developed geospatial visualization and mapping systems using OpenGL and CesiumJS, including rendering pipelines, coordinate transforms, and performance optimization for large datasets
- Profiled and debugged a 30+ thread multithreaded application in Windows and Linux environments using Visual Studio debugger, VS Code, GDB, Valgrind
- Modernized legacy C++ codebase to modern C++ standards (C++11/14/17 style), improving memory safety, performance, and maintainability
- Performed performance tuning including CPU/memory profiling and optimization
- Contributed to technical documentation, system design notes, and internal developer onboarding materials for complex multi-component software architecture
- Leveraged AI-assisted development tools (Claude) to accelerate debugging, refactoring, and code understanding across large legacy C++ systems

Hardware / Systems Engineer — Dexlabs (dexlabs.co)

2016 – 2026

- Designed and developed embedded systems from concept to production, including schematic design, PCB layout, firmware development, mechanical enclosure design, prototyping, and hardware validation
- Engineered STM32 (ARM Cortex-M) based PCBs, including full hardware bring-up, debugging, validation, and application of DFM/DFA principles
- Developed embedded firmware in C/C++ using bare-metal, HAL, and CMSIS-style abstractions for memory-constrained, real-time systems, applying interrupt-driven architectures, state machines, and low-power/performance optimization techniques
- Implemented and debugged peripheral interfaces including UART, SPI, I2C, CAN, USB, PWM, ADC/DAC,

timers, interrupts, and DMA
- Performed PCB bring-up, hardware/software integration, validation, and system-level debugging using oscilloscopes, logic analyzers, SWD/J-Link, and GDB-based workflows
- Designed and developed custom input devices including USB HID peripherals (e.g., computer mice) and force-feedback game controllers, including firmware, protocol implementation, and real-time actuator control; reverse engineered Bluetooth protocols (e.g., Nintendo Switch Joy-Con) for third-party compatibility
- Managed component sourcing, BOM optimization, and supply chain workflows using vendors such as DigiKey, JLCPCB, and PCBWay, including coordination with PCB fabrication and assembly partners
- Operated and maintained 3D printing systems for rapid prototyping and iterative hardware development
- Created technical documentation including schematics, design notes, validation procedures, and manufacturing documentation

Engineering Internship — Liburdi Automation (www.liburdiautomation.com)

Aug 2008 – Sep 2009

- Designed and built a rugged electronic test system for the mainboard in a welding machine, including full CAD documentation.
- Developed embedded C software on a PIC microcontroller for a robotic control system (3-axis motor control)
- Wrote control software in Visual Basic for a motor controller attached via RS-232
- Supported engineering and production teams with troubleshooting, validation, and system integration
- Worked in an industrial robotics environment supporting aerospace and heavy industry applications

Education

Bachelor of Engineering — Computer Engineering (Co-Op)

McMaster University, Ontario, Canada
2005 – 2011

Skills and experience

Hardware Skills

- Developing bare-metal embedded firmware in C/C++ for STM32 ARM Cortex and PIC microcontrollers, using STM32Cube, Arduino, and MPLAB toolchains with ST-Link and J-Link debugging tools, while designing for real-time, memory, and power constraints
- Designing schematics and 2-layer/4-layer PCB layouts using KiCad and Eagle, including review for correctness and manufacturability
- Developing prototypes based on Arduino development boards, including ESP32, for rapid prototyping
- Designing firmware using state machine-based architecture
- Performing embedded hardware and firmware debugging and validation using oscilloscopes and logic analyzers for signal measurement and protocol analysis
- Managing infrastructure including Hyper-V virtual machines (Ubuntu-based) and small networks of servers, switches, access points, and PoE camera systems
- Designing analog filters, high-order filters, and active filters
- Working with serial communication interfaces: RS-232, RS-485, SPI, CAN, I2C, UART
- Performing hand soldering for prototyping, repair, and PCB rework
- Designing SMPS power supplies, including component selection and PCB layout
- Following ESD-safe handling procedures during hardware work
- Performing reverse engineering of PCB assemblies, including schematic capture, component identification, signal analysis using logic analyzers and oscilloscopes, and circuit tracing with multimeters
- Developing protection and charging systems for lithium batteries
- Developing embedded systems with Bluetooth, BLE, and Wi-Fi connectivity and TCP/IP networking with REST and MQTT interfaces

Software Skills

- Developing desktop and embedded software (Windows, Linux, bare metal) in C and C++, building and debugging using MSVC, GCC, and VS Code toolchains
- Developing and debugging web applications and backend software using JavaScript, Node.js, REST APIs, React, Next.js, TypeScript, jQuery
- Using Git for version control and collaboration
- Using Python, CMake, shell scripts, and Lua for build automation and development tooling
- Collaborating across multidisciplinary teams to design, implement, and deliver complex software features in large-scale production systems
- Creating complex SQL schemas with relationships and indexes and performance tuning queries on tables with millions of rows
- Deploying and integrating systems on AWS (S3, EC2, Route 53)
- Bridging backend and frontend systems — connecting C++ services running in Docker containers and SQL databases to JavaScript-based web frontends over REST and WebSocket
- Developing and deploying backend services using Docker-based infrastructure, Docker Compose, Swarm
- Designing maintainable software components following established coding standards with emphasis on performance, scalability, and code quality
- Performing peer code reviews focused on code quality, maintainability, and standards compliance
- Working extensively with modern C++ practices including RAII, smart pointers, STL containers, and concurrency primitives
- Troubleshooting and resolving complex concurrency, memory management, and runtime stability issues in multithreaded systems
- Developing high-performance rendering and visualization features for large geospatial and telemetry datasets
- Supporting migration and modernization efforts for legacy software systems while maintaining backward compatibility

Prototyping and Manufacturing

- Using FFF 3D printing for rapid prototyping of enclosures and mechanical systems
- Designing sheet metal components for mechanical assemblies
- Sourcing electronic and mechanical components for prototype designs and repairs (e.g., DigiKey, LCSC, McMaster-Carr)
- Designing PCBs and coordinating with PCB fabrication and assembly vendors (e.g., PCBWay, JLCPCB) to fabricate and assemble parts, prototypes, and products
- Applying principles of design for manufacturing, design for assembly, design for 3D printing
- Performing prototype validation and functional testing using custom-designed hardware test fixtures
- Supporting prototype-to-production development for embedded hardware products

Technical Skills

- Applying structured problem solving to engineering challenges
- Applying system-level thinking across cross-disciplinary engineering projects
- Working with multidisciplinary engineering teams on integrated systems
- Producing technical documentation for software and hardware systems
- Participating in collaborative software and hardware development projects
- Working across cross-disciplinary engineering domains including hardware, firmware, software, and mechanical systems
- Performing failure analysis on hardware, software, embedded and mechanical systems
- Developing systems based on technical and operational requirements
- Leveraging AI-assisted development workflows to accelerate software development
- Preparing patents for the USPTO, working with inventors, lawyers, draftspeople, and patent examiners to see a patent through from inception to acceptance